

Assignment No. 7

VPC Peering Between Two AWS VPCs

Objective:

The objective of this assignment is to establish secure communication between two different Amazon Virtual Private Clouds (VPCs) using **AWS VPC Peering**. Two VPCs are created with separate CIDR blocks, each containing an EC2 instance. A VPC Peering Connection is established between them, route tables are updated, and connectivity is verified using the ping command between the EC2 instances.

Problem Statement:

Organizations often deploy applications in multiple VPCs to isolate workloads for security and management purposes. However, these VPCs cannot communicate with each other by default.

AWS VPC Peering provides a private network connection between two VPCs, allowing EC2 instances to communicate using private IP addresses without traversing the public internet.

AWS Services Used

- Amazon VPC
- Amazon EC2
- VPC Peering Connection
- Internet Gateway
- Public Subnets
- Route Tables
- Security Groups
- Network ACL
- SSH
- ICMP (Ping)

Network Design:

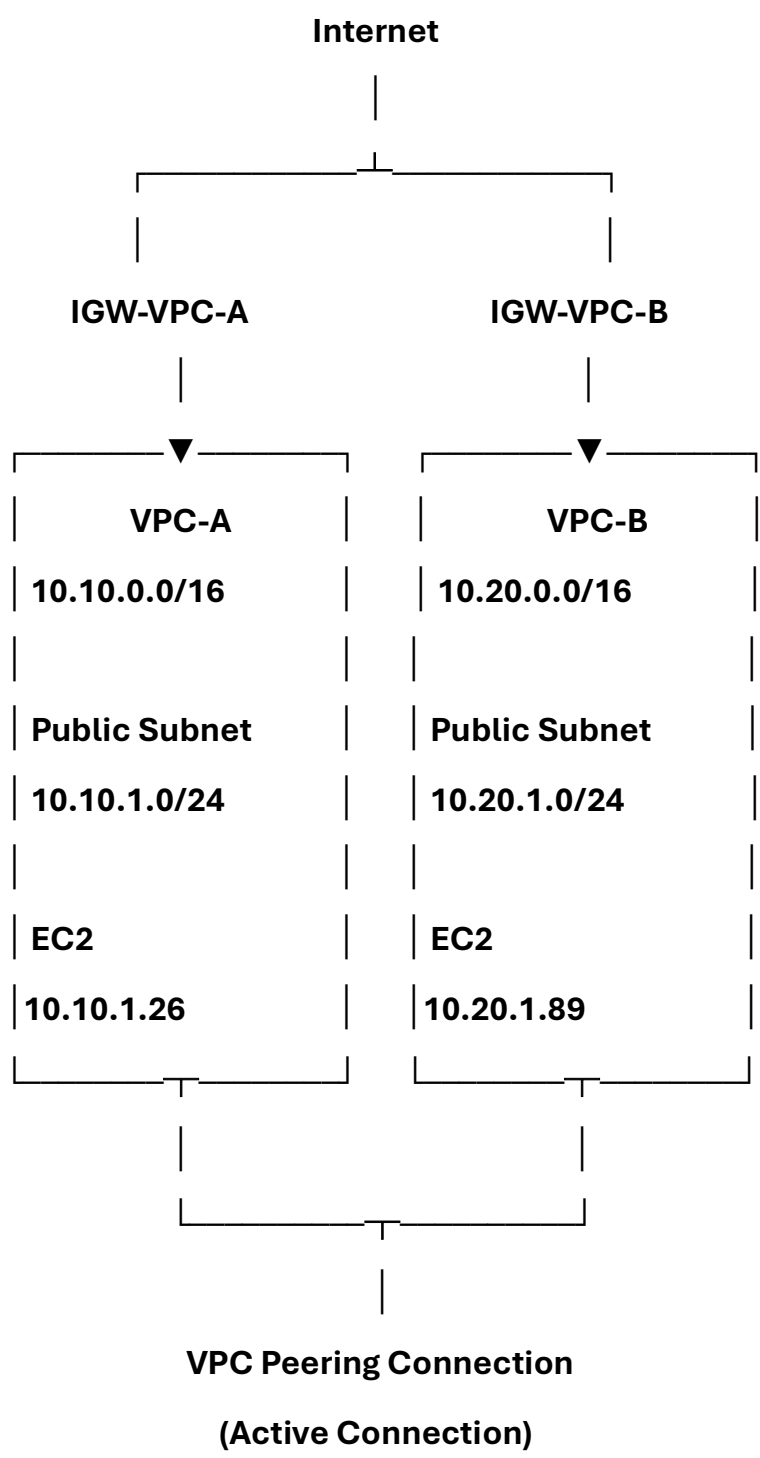
VPC-A:

Property	Value
CIDR Block	10.10.0.0/16
Public Subnet	10.10.1.0/24
EC2 Private IP	10.10.1.26

VPC-B:

Property	Value
CIDR Block	10.20.0.0/16
Public Subnet	10.20.1.0/24
EC2 Private IP	10.20.1.89

Architecture:



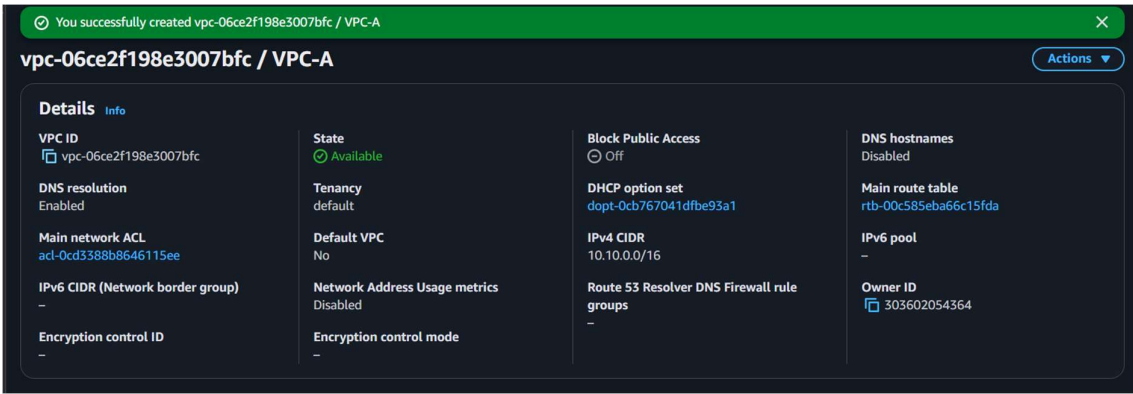
Implementation Steps

Step 1: Create VPC-A

A Virtual Private Cloud named **VPC-A** was created with the CIDR block **10.10.0.0/16**. This VPC serves as the first isolated virtual network where the first EC2 instance will be deployed.

Configuration

- VPC Name : **VPC-A**
- IPv4 CIDR : **10.10.0.0/16**
- DNS Resolution : Enabled
- DNS Hostnames : Disabled (Default)

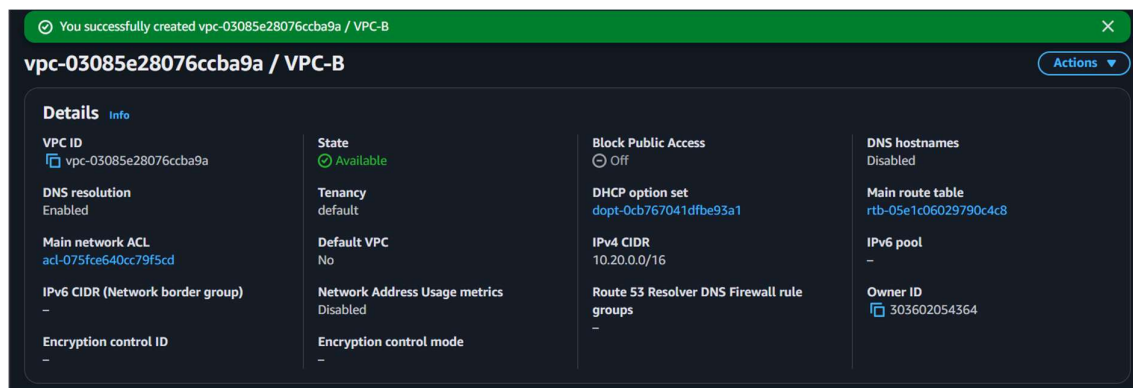


Step 2: Create VPC-B

A second Virtual Private Cloud named **VPC-B** was created with the CIDR block **10.20.0.0/16**. This VPC acts as the second isolated network that will communicate with VPC-A using VPC Peering.

Configuration

- VPC Name : **VPC-B**
- IPv4 CIDR : **10.20.0.0/16**
- DNS Resolution : Enabled
- DNS Hostnames : Disabled (Default)



Step 3: Create Public Subnets

A public subnet was created inside each VPC to host the EC2 instances.

VPC-A Public Subnet

- CIDR Block : **10.10.1.0/24**
- Auto Assign Public IPv4 : **Enabled**

The screenshot shows the AWS Subnets console. At the top, there's a search bar and a table of subnets. The table has columns: Name, Subnet ID, State, VPC, Block Public Access, and IPv4 CIDR. The 'VPC-A-Public-Subnet' is selected. Below the table, the details for 'subnet-062530454ce4ee8eb / VPC-A-Public-Subnet' are shown. The details are organized into four columns: Subnet ID, Subnet ARN, State, and Block Public Access. The Subnet ID is 'subnet-062530454ce4ee8eb'. The Subnet ARN is 'arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-062530454ce4ee8eb'. The State is 'Available'. The Block Public Access is 'Off'. Other details include: IPv4 CIDR: 10.10.1.0/24, Availability Zone: ap-south-1a, Network ACL: -, Auto-assign customer-owned IPv4 address: No, Available IPv4 addresses: 251, Network border group: ap-south-1, Default subnet: No, Customer-owned IPv4 pool: -, IPv6 CIDR: -, VPC: vpc-06ce2f198e3007bfc | VPC-A, Auto-assign public IPv4 address: Yes, Outpost ID: -, Hostname type: -, IPv6 CIDR association ID: -, Route table: -, Auto-assign IPv6 address: No, IPv4 CIDR reservations: -, and Resource name DNS A record: -.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-04f4ef3b23eb11465	Available	vpc-055fefadc90488fe4	Off	172.31.16.0/2
-	subnet-0d8f9020d97c12a3a	Available	vpc-055fefadc90488fe4	Off	172.31.32.0/2
VPC-A-Public-Subnet	subnet-062530454ce4ee8eb	Available	vpc-06ce2f198e3007bfc VPC-A	Off	10.10.1.0/24

subnet-062530454ce4ee8eb / VPC-A-Public-Subnet

Subnet ID	Subnet ARN	State	Block Public Access
subnet-062530454ce4ee8eb	arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-062530454ce4ee8eb	Available	Off

IPv4 CIDR: 10.10.1.0/24

Availability Zone: ap-south-1a

Network ACL: -

Auto-assign customer-owned IPv4 address: No

Available IPv4 addresses: 251

Network border group: ap-south-1

Default subnet: No

Customer-owned IPv4 pool: -

IPv6 CIDR: -

VPC: vpc-06ce2f198e3007bfc | VPC-A

Auto-assign public IPv4 address: Yes

Outpost ID: -

Hostname type: -

IPv6 CIDR association ID: -

Route table: -

Auto-assign IPv6 address: No

IPv4 CIDR reservations: -

Resource name DNS A record: -

VPC-B Public Subnet

- CIDR Block : **10.20.1.0/24**
- Auto Assign Public IPv4 : **Enabled**

The screenshot shows the AWS Subnets console. At the top, there's a search bar and a table of subnets. The table has columns: Name, Subnet ID, State, VPC, Block Public Access, and IPv4 CIDR. The 'VPC-B-Public-Subnet' is selected. Below the table, the details for 'subnet-0354df6576822d848 / VPC-B-Public-Subnet' are shown. The details are organized into four columns: Subnet ID, Subnet ARN, State, and Block Public Access. The Subnet ID is 'subnet-0354df6576822d848'. The Subnet ARN is 'arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-0354df6576822d848'. The State is 'Available'. The Block Public Access is 'Off'. Other details include: IPv4 CIDR: 10.20.1.0/24, Availability Zone: ap-south-1a, Network ACL: -, Auto-assign customer-owned IPv4 address: No, Available IPv4 addresses: 251, Network border group: ap-south-1, Default subnet: No, Customer-owned IPv4 pool: -, IPv6 CIDR: -, VPC: vpc-03085e28076ccba9a | VPC-B, Auto-assign public IPv4 address: Yes, Outpost ID: -, Hostname type: -, IPv6 CIDR association ID: -, Route table: -, Auto-assign IPv6 address: No, IPv4 CIDR reservations: -, and Resource name DNS A record: -.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-0d8f9020d97c12a3a	Available	vpc-055fefadc90488fe4	Off	172.31.32.0/2
VPC-A-Public-Subnet	subnet-062530454ce4ee8eb	Available	vpc-06ce2f198e3007bfc VPC-A	Off	10.10.1.0/24
VPC-B-Public-Subnet	subnet-0354df6576822d848	Available	vpc-03085e28076ccba9a VPC-B	Off	10.20.1.0/24

subnet-0354df6576822d848 / VPC-B-Public-Subnet

Subnet ID	Subnet ARN	State	Block Public Access
subnet-0354df6576822d848	arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-0354df6576822d848	Available	Off

IPv4 CIDR: 10.20.1.0/24

Availability Zone: ap-south-1a

Network ACL: -

Auto-assign customer-owned IPv4 address: No

Available IPv4 addresses: 251

Network border group: ap-south-1

Default subnet: No

Customer-owned IPv4 pool: -

IPv6 CIDR: -

VPC: vpc-03085e28076ccba9a | VPC-B

Auto-assign public IPv4 address: Yes

Outpost ID: -

Hostname type: -

IPv6 CIDR association ID: -

Route table: -

Auto-assign IPv6 address: No

IPv4 CIDR reservations: -

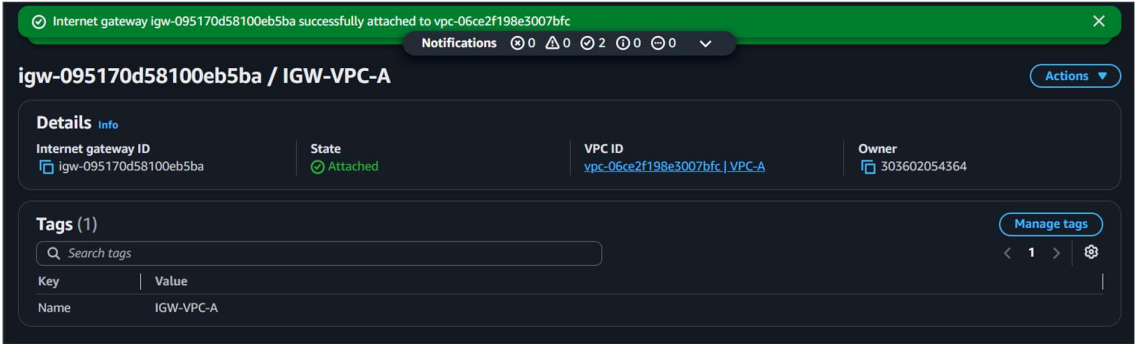
Resource name DNS A record: -

Step 4: Create and Attach Internet Gateways

To provide internet connectivity for the EC2 instances, an Internet Gateway was created and attached to each VPC.

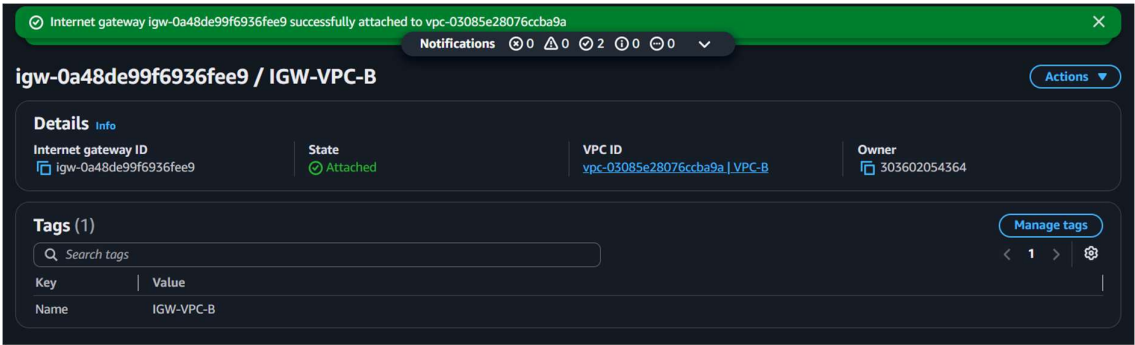
Internet Gateway for VPC-A

- Created IGW
- Attached to **VPC-A**



Internet Gateway for VPC-B

- Created IGW
- Attached to **VPC-B**



Step 5: Configure Route Table for VPC-A

A custom route table was created and associated with the public subnet of VPC-A.

The following routes were added:

Destination	Target
10.10.0.0/16	Local
0.0.0.0/0	Internet Gateway

This allows EC2 instances in VPC-A to access the internet.

You have successfully updated subnet associations for rtb-0417b9c36b1d5a33a / RT-VPC-A.

rtb-0417b9c36b1d5a33a / RT-VPC-A

Details

Route table ID
rtb-0417b9c36b1d5a33a

VPC
vpc-06ce2f198e3007bfc | VPC-A

Main

No

Owner ID
303602054364

Explicit subnet associations

subnet-062530454ce4ee8eb / VPC-A-Public-Subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-095170d58100eb5ba	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table

You have successfully updated subnet associations for rtb-0417b9c36b1d5a33a / RT-VPC-A.

rtb-0417b9c36b1d5a33a / RT-VPC-A

Details

Route table ID
rtb-0417b9c36b1d5a33a

VPC
vpc-06ce2f198e3007bfc | VPC-A

Main

No

Owner ID
303602054364

Explicit subnet associations

subnet-062530454ce4ee8eb / VPC-A-Public-Subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
VPC-A-Public-Subnet	subnet-062530454ce4ee8eb	10.10.1.0/24	-

Subnets without explicit associations (0)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
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Step 6: Configure Route Table for VPC-B

Similarly, a route table was created for VPC-B and associated with its public subnet.

The following routes were configured:

Destination	Target
10.20.0.0/16	Local
0.0.0.0/0	Internet Gateway

This enables internet connectivity for EC2 instances inside VPC-B.

Route tables (1/5) Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-05e1c06029790c4c8	-	-	Yes	vpc-03085e28076ccba9a VPC
☐	-	rtb-00c585eba66c15fda	-	-	Yes	vpc-06ce2f198e3007bfc VPC
☒	RT-VPC-B	rtb-0b04f3134ee572b7f	subnet-0354df6576822d...	-	No	vpc-03085e28076ccba9a VPC

rtb-0b04f3134ee572b7f / RT-VPC-B

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0a48de99f6936fee9	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

Route tables (1/5) Info

Last updated 1 minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-05e1c06029790c4c8	-	-	Yes	vpc-03085e28076ccba9a VPC
☐	-	rtb-00c585eba66c15fda	-	-	Yes	vpc-06ce2f198e3007bfc VPC
☒	RT-VPC-B	rtb-0b04f3134ee572b7f	subnet-0354df6576822d...	-	No	vpc-03085e28076ccba9a VPC

rtb-0b04f3134ee572b7f / RT-VPC-B

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
VPC-B-Public-Subnet	subnet-0354df6576822d848	10.20.1.0/24	-

Step 7: Launch EC2 Instances

One Ubuntu EC2 instance was launched inside each VPC. Each instance was deployed in its respective public subnet with an automatically assigned public IPv4 address, allowing SSH access from the internet.

EC2 Instance in VPC-A

Configuration:

- Instance Name: **EC2-VPC-A**
- AMI: **Ubuntu Server**
- Instance Type: **t3.micro**
- VPC: **VPC-A**
- Subnet: **VPC-A-Public-Subnet**
- Private IP: **10.10.1.26**
- Public IPv4: Auto Assigned

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, the 'Instances' section shows a table with one instance, 'EC2-VPC-A', which is in a 'Running' state. Below the table, the detailed configuration for instance 'i-09f0a7e7729f1dfb7' is shown. The instance is a 't3.micro' type, located in the 'ap-south-1a' availability zone. It has a public IPv4 address of 3.110.190.147 and a private IP address of 10.10.1.26. The instance is running on the 'Ubuntu Server' AMI. The VPC ID is 'vpc-06ce2f198e3007bfc (VPC-A)'. The instance has an auto-assigned public IP address and a private IP address. The instance is running on the 't3.micro' instance type. The instance is running on the 'Ubuntu Server' AMI. The instance is running on the 't3.micro' instance type. The instance is running on the 'Ubuntu Server' AMI.

Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
EC2-VPC-A	Running	t3.micro	3/3 checks passed	ap-south-1a	-

i-09f0a7e7729f1dfb7 (EC2-VPC-A)		
Instance ID i-09f0a7e7729f1dfb7	Public IPv4 address 3.110.190.147 open address	Private IPv4 addresses 10.10.1.26
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-10-1-26.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-10-1-26.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.110.190.147 [Public IP]	VPC ID vpc-06ce2f198e3007bfc (VPC-A)	

EC2 Instance in VPC-B

Configuration:

- Instance Name: **EC2-VPC-B**
- AMI: **Ubuntu Server**
- Instance Type: **t3.micro**
- VPC: **VPC-B**
- Subnet: **VPC-B-Public-Subnet**
- Private IP: **10.20.1.89**
- Public IPv4: Auto Assigned

Instances (1/2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	EC2-VPC-A	i-09f0a7e7729f1dfb7	Running	t3.micro	3/3 checks passed	ap-south-1a	-	3
☒	EC2-VPC-B	i-04ed1b78a7bfd0393	Running	t3.micro	3/3 checks passed	ap-south-1a	-	1

i-04ed1b78a7bfd0393 (EC2-VPC-B)

Instance ID

i-04ed1b78a7bfd0393

IPv6 address

-

Hostname type

IP name: ip-10-20-1-89.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

13.232.99.210 [Public IP]

Public IPv4 address

13.232.99.210 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-20-1-89.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-03085e28076ccba9a (VPC-B)

Private IPv4 addresses

10.20.1.89

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Step 8: Create VPC Peering Connection

A VPC Peering Connection was created to establish private network communication between VPC-A and VPC-B. After creating the request, it was accepted, changing the status of the connection to **Active**.

VPC Peering Details

Property	Value
Name	VPC-A-to-VPC-B
Requester VPC	VPC-A
Acceptor VPC	VPC-B
Status	Active

The active VPC Peering Connection enables traffic exchange between the two VPCs over AWS's private network without using the public internet.

🔍 Your VPC peering connection (pcx-0032fdffd882c6f0a / VPC-A-to-VPC-B) has been established.

To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.

Info

[Modify my route tables now](#)

✕

Peering connections (1/1) Info

🔍

🔄

[Actions](#)

[Create peering connection](#)

<
1
>
⚙️

Name	Peering connection ID	Status	Requester VPC	Acceptor VPC
🔍 VPC-A-to-VPC-B	pcx-0032fdffd882c6f0a	✔️ Active	vpc-06ce2f198e3007bfc / VPC-A	vpc-03085e28076ccba9a / VPC-B

⏪

⏩

Details

Requester owner ID

🔍
303602054364

Peering connection ID

🔍
pcx-0032fdffd882c6f0a

Status

✔️
Active

Expiration time

🔍
-

Requester owner ID

🔍
303602054364

Requester VPC

🔍
vpc-06ce2f198e3007bfc / VPC-A

Requester CIDRs

🔍
10.10.0.0/16

Requester Region

🔍
Mumbai (ap-south-1)

VPC Peering connection ARN

🔍
arn:aws:ec2:ap-south-1:303602054364:vpc-peering-connection/pcx-0032fdffd882c6f0a

Acceptor VPC

🔍
vpc-03085e28076ccba9a / VPC-B

Acceptor CIDRs

🔍
10.20.0.0/16

Acceptor Region

🔍
Mumbai (ap-south-1)

Step 9: Update Route Tables

After the VPC Peering Connection became active, both custom route tables were updated to direct traffic destined for the peer VPC through the peering connection.

Route Table of VPC-A

The following route was added:

Destination	Target
10.20.0.0/16	VPC Peering Connection

Final Route Table:

Destination	Target
10.10.0.0/16	Local
10.20.0.0/16	VPC Peering Connection
0.0.0.0/0	Internet Gateway

Updated routes for rtb-0417b9c36b1d5a33a / RT-VPC-A successfully
Details

rtb-0417b9c36b1d5a33a / RT-VPC-AActions

Detailsinfo

Route table ID
rtb-0417b9c36b1d5a33a

VPC
vpc-06ce2f198e3007bfc | VPC-A

Main
No

Owner ID
303602054364

Explicit subnet associations
subnet-062530454ce4ee8eb / VPC-A-Public-Subnet

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTagsRelated resources

Routes (3)BothEdit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-095170d58100eb5ba	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table
10.20.0.0/16	pcx-0032fd9df882c6f0a	Active	No	Create Route

Route Table of VPC-B

The following route was added:

Destination	Target
10.10.0.0/16	VPC Peering Connection

Final Route Table:

Destination	Target
10.20.0.0/16	Local
10.10.0.0/16	VPC Peering Connection
0.0.0.0/0	Internet Gateway

Updated routes for rtb-0b04f3134ee572b7f / RT-VPC-B successfully

Details

rtb-0b04f3134ee572b7f / RT-VPC-B

Actions

Details

Info

Route table ID

rtb-0b04f3134ee572b7f

VPC

vpc-03085e28076ccba9a | VPC-B

Main

No

Owner ID

303602054364

Explicit subnet associations

subnet-0354df6576822d848 / VPC-B-Public-Subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (3)

BothEdit routes

Filter routes

<1>

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0a48de99f6936fee9	Active	No	Create Route
10.10.0.0/16	pcc-0032fd82c6f0a	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

Step 10: Verify Network ACL Configuration

The default Network ACLs associated with both public subnets were verified. Since the default Network ACL allows all inbound and outbound traffic, no additional modifications were required.

Network ACL Verification - VPC-A

Verified Rules:

- Inbound: Allow All Traffic
- Outbound: Allow All Traffic

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

Name

Network ACL ID

Associated with

Default

VPC ID

Inbound

acl-075fce640cc79f5cd

subnet-0354df6576822d848 / VPC-B-Public-S...

Yes

vpc-03085e28076ccba9a / VPC-B

2 Inbound

acl-0a8ed10d4bb027a60

3 Subnets

Yes

vpc-055fefadc90488fe4

2 Inbound

acl-0cd3388b8646115ee

subnet-062530454ce4ee8eb / VPC-A-Public-Su...

Yes

vpc-06ce2f198e3007bfc / VPC-A

2 Inbound

acl-0cd3388b8646115ee

Details

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (2)

Edit inbound rules

Filter inbound rules

Rule number

Type

Protocol

Port range

Source

Allow/Deny

100

All traffic

All

All

0.0.0.0/0

Allow

*

All traffic

All

All

0.0.0.0/0

Deny

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

Name

Network ACL ID

Associated with

Default

VPC ID

Inbound

acl-075fce640cc79f5cd

subnet-0354df6576822d848 / VPC-B-Public-S...

Yes

vpc-03085e28076ccba9a / VPC-B

2 Inbound

acl-0a8ed10d4bb027a60

3 Subnets

Yes

vpc-055fefadc90488fe4

2 Inbound

acl-0cd3388b8646115ee

subnet-062530454ce4ee8eb / VPC-A-Public-Su...

Yes

vpc-06ce2f198e3007bfc / VPC-A

2 Inbound

acl-0cd3388b8646115ee

Details

Inbound rules

Outbound rules

Subnet associations

Tags

Outbound rules (2)

Edit outbound rules

Filter outbound rules

Rule number

Type

Protocol

Port range

Destination

Allow/Deny

100

All traffic

All

All

0.0.0.0/0

Allow

*

All traffic

All

All

0.0.0.0/0

Deny

Network ACL Verification - VPC-B

Verified Rules:

- Inbound: Allow All Traffic
- Outbound: Allow All Traffic

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

Actions

Create network ACL

< 1 >

	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☒	-	acl-075fce640cc79f5cd	subnet-0354df6576822d848 / VPC-B-Public-S...	Yes	vpc-03085e28076ccb9a / VPC-B	2 Inbound
☐	-	acl-0a8ed10d4bb027a60	3 Subnets	Yes	vpc-055fefadc90488fe4	2 Inbound
☐	-	acl-0cd3388b8646115ee	subnet-062530454ce4ee8eb / VPC-A-Public-Su...	Yes	vpc-06ce2f198e3007bfc / VPC-A	2 Inbound

acl-075fce640cc79f5cd

Details

Inbound rules

Outbound rules

Subnet associations

Tags

Inbound rules (2)

Edit inbound rules

Filter inbound rules

< 1 >

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Network ACLs (1/3) Info

Find Network ACLs by attribute or tag

Actions

Create network ACL

< 1 >

	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☒	-	acl-075fce640cc79f5cd	subnet-0354df6576822d848 / VPC-B-Public-S...	Yes	vpc-03085e28076ccb9a / VPC-B	2 Inbound
☐	-	acl-0a8ed10d4bb027a60	3 Subnets	Yes	vpc-055fefadc90488fe4	2 Inbound
☐	-	acl-0cd3388b8646115ee	subnet-062530454ce4ee8eb / VPC-A-Public-Su...	Yes	vpc-06ce2f198e3007bfc / VPC-A	2 Inbound

acl-075fce640cc79f5cd

Details

Inbound rules

Outbound rules

Subnet associations

Tags

Outbound rules (2)

Edit outbound rules

Filter outbound rules

< 1 >

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Step 11: Verify Connectivity Between VPCs

After configuring the VPC Peering Connection and updating both route tables, connectivity between the EC2 instances was verified using the private IP address of the instance in VPC-B.

The EC2 instance in VPC-A was accessed using SSH, and an ICMP (ping) request was sent to the private IP address of the EC2 instance in VPC-B.

SSH Connection to EC2-VPC-A

The EC2 instance was successfully accessed from the local machine using the SSH key pair.

Command Used:

```
ssh -i aws-keypair.pem ubuntu@3.110.190.147
```

The login was successful, confirming that:

- SSH Security Group rule was correctly configured.
- Internet Gateway and Route Table were functioning correctly.
- Public IP assignment was working properly.

```
PS C:\Users\YASH\Downloads> ssh -i aws-keypair.pem ubuntu@3.110.190.147
The authenticity of host '3.110.190.147 (3.110.190.147)' can't be established.
ED25519 key fingerprint is SHA256:BRf9iAnTZzL9ziNOwR7DMwOjWxGxuWEgiQyI7wkTV4Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.110.190.147' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul 14 15:25:28 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.9% of 6.61GB Processes:    115
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.10.1.26

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-10-1-26:~$ |
```

Ping Test Using Private IP

From EC2-VPC-A, the following command was executed:

```
ping 10.20.1.89
```

The ping replies were received successfully with **0% packet loss**, confirming that:

- VPC Peering Connection was working correctly.
- Route Tables were properly configured.
- Security Groups allowed ICMP traffic.
- Private communication between both VPCs was established successfully.

```
ubuntu@ip-10-10-1-26:~$ ping 10.20.1.89
PING 10.20.1.89 (10.20.1.89) 56(84) bytes of data.
64 bytes from 10.20.1.89: icmp_seq=1 ttl=64 time=0.796 ms
64 bytes from 10.20.1.89: icmp_seq=2 ttl=64 time=0.169 ms
64 bytes from 10.20.1.89: icmp_seq=3 ttl=64 time=0.344 ms
64 bytes from 10.20.1.89: icmp_seq=4 ttl=64 time=0.331 ms
64 bytes from 10.20.1.89: icmp_seq=5 ttl=64 time=0.859 ms
64 bytes from 10.20.1.89: icmp_seq=6 ttl=64 time=0.352 ms
64 bytes from 10.20.1.89: icmp_seq=7 ttl=64 time=0.658 ms
64 bytes from 10.20.1.89: icmp_seq=8 ttl=64 time=0.330 ms
64 bytes from 10.20.1.89: icmp_seq=9 ttl=64 time=0.395 ms
64 bytes from 10.20.1.89: icmp_seq=10 ttl=64 time=0.315 ms
64 bytes from 10.20.1.89: icmp_seq=11 ttl=64 time=0.334 ms
^C
--- 10.20.1.89 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10251ms
rtt min/avg/max/mdev = 0.169/0.443/0.859/0.211 ms
ubuntu@ip-10-10-1-26:~$ |
```

Challenges Faced

- Initial SSH connection to EC2-VPC-A failed because the local system's public IP had changed, requiring the SSH rule in the Security Group to be updated.
- The first EC2 instance was launched before configuring all networking components, so a new instance was launched after completing the network setup.
- Route Tables were updated after creating the VPC Peering Connection to enable communication between the VPCs.
- Connectivity was verified only after confirming the correct subnet association, Internet Gateway attachment, Security Group configuration, and routing.

Conclusion:

In this assignment, two independent Virtual Private Clouds (VPC-A and VPC-B) were created with separate CIDR blocks and public subnets. Internet connectivity was enabled using Internet Gateways, and custom Route Tables were configured for proper network routing. Ubuntu EC2 instances were launched in each VPC, and a VPC Peering Connection was established to enable private communication between them. Finally, connectivity was verified successfully using a ping test over private IP addresses, demonstrating successful implementation of secure inter-VPC communication using AWS VPC Peering.